# Quantum Tunneling in Biology

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## 1 Introduction

Biology is traditionally defined as the scientific study of life and living organisms, encompassing their structure, function, and growth. In contrast, quantum mechanics is the branch of physics dedicated to the behavior of matter and energy at the atomic and subatomic levels. At first glance, these fields appear uncorrelated and non-overlapping; living organisms are macroscopic entities, significantly larger than atoms or molecules, leading to the assumption that quantum effects should be negligible at the biological scale.

However, this classical view is increasingly being challenged. It has become evident that certain organisms do not merely tolerate quantum mechanical phenomena but benefit from them to the extent of functional necessity. A prime macroscopic example is the gecko, which exploits Van der Waals forces—a quantum phenomenon—to generate adhesion forces allowing it to stick to various surfaces. Beyond adhesion, nature appears to exploit quantum effects to optimize the efficiency of fundamental biological processes.

The inadequacy of classical mechanics becomes apparent when modeling complex biological systems. For instance, in photosynthesis, classical models predict significant energy dissipation and slower reaction times, contradicting experimental observations of near-perfect efficiency. Similarly, classical transition state theory fails to fully explain the catalytic proficiency of certain enzymes or the mechanics of spontaneous DNA mutations.

This paper explores the premise that quantum tunneling and coherence are fundamental processes for living organisms. We will examine three distinct case studies: the energy transfer phase of photosynthesis, the catalytic activity of enzymes, and the stochastic nature of DNA mutations. In doing so, we aim to demonstrate that biological systems are not strictly classical but operate as open quantum systems that harness environmental noise to facilitate quantum transport and chemical reactivity.

## 2 Study Case I: Photosynthesis

Photosynthesis is the basis of life. It absorbs $CO_2$ to produce $O_2$, allowing us heterotrophs to live, while producing sugar for the organism to live and grow.

This process is mainly separated into two phases: the light reactions and the dark reactions. In the light reactions the system absorbs water and light to produce oxygen, adenosine triphosphate (ATP) and NADPH, two molecules required for the dark reactions, the so-called Calvin cycle, which produce sugar for the plant to consume.

As for any process in living organisms, natural selection has made sure to optimize any process required for the survival of the organism, and photosynthesis is no exception. Sunlight is absorbed by the chlorophyll pigments, arranged in a structure called antenna, to then pass this energy to the reaction center, where the energy is used to turn adenosine diphosphate (ADP) into ATP.

This energy transfer is surprisingly optimal, reaching an efficiency of more than 99%. It has also been seen that this process is extremely fast. It takes around 10 picoseconds ($10^{-11}$s) to absorb light

and use its energy in the reaction center.

When treating this process in a classical way, the models created to simulate this system have inconsistencies with reality. For instance, the classical models show that the energy transfer process should have a maximum efficiency of around 70%, losing a considerable amount of energy due to dissipation. The models also show that the process would be much slower, in the order of magnitude of picoseconds ($10^{-9}$s).

Classical Förster resonance energy transfer (FRET) models assume that energy migrates through the complex via incoherent, step-wise hopping between independent pigments. This implies that each transfer event is probabilistic, thermally activated, and occurs on nanosecond time scales. Such models also predict substantial losses due to the random walk nature of the pathway: excitons should frequently revisit pigments, dissipate energy to the environment, and fail to reach the reaction center before relaxation. These predictions conflict with experimental observations, which show near-unity quantum yields and picosecond-scale transport. The mismatch strongly suggests that additional non-classical transport mechanisms must be considered.

This inconsistency shows that the process might not be using only classical mechanics. Considering the scale of the processes, it is reasonable to consider quantum mechanics having some effects on the system.

In photosynthetic complexes, the absorbed photon does not travel as a particle of light between pigments. Instead, its energy is converted into an electronic excitation called an exciton. An exciton is a quasiparticle formed by a bound electron–hole pair, and it behaves according to quantum mechanical principles. Importantly, the exciton is not confined to a single chlorophyll molecule. Because the pigments are embedded in a dense protein matrix with nanometer-level separation, their electronic orbitals overlap sufficiently to allow the exciton to delocalize over several pigments. This delocalization is the physical basis for both coherence and interference within the light-harvesting complex, and it explains why standard classical hopping models underestimate the speed and efficiency of energy transfer.

## 2.1 Quantum coherence

The main problem of the classical models is that they treat chlorophyll pigments as well-defined independent units. If this was the case, a considerable portion of the energy would be lost during the transfer of energy from one pigment to another, and the process of searching for the reaction center is very slow, as it is basically a random walk through the pigments.

Experimental confirmation of quantum coherence in photosynthetic complexes emerged from ultrafast two-dimensional electronic spectroscopy (2DES) experiments, notably those performed on the Fenna–Matthews–Olson (FMO) complex. These experiments revealed oscillatory signals in the electronic spectra that persist for several hundred femtoseconds, far longer than previously expected for warm biological environments. Such oscillations correspond to coherent superpositions of excitonic states. Although the coherence lifetime is short compared to the total energy-transfer time, these coherent intervals are sufficient to bias the exciton's movement toward the reaction center, improving both speed and trapping efficiency relative to incoherent hopping.

Quantum coherence gives a possible explanation for the slow pace of classical models. Quantum coherence is the ability of a system to exist in a superposition of quantum states, which enable phenomena such as quantum interference and delocalization.

As Jianshu Cao and Robert J. Silbey explain in their paper Optimization of Exciton Trapping in Energy Transfer Processes (2009), photosynthesis exploits the delocalization phenomena to make the energy transfer faster. Instead of searching with only one path at a time (which is what a classical random walk would do), delocalization allows the energy to be simultaneously on several nearby pigments, therefore allowing the exploration to be on multiple paths at once, reaching the reaction center

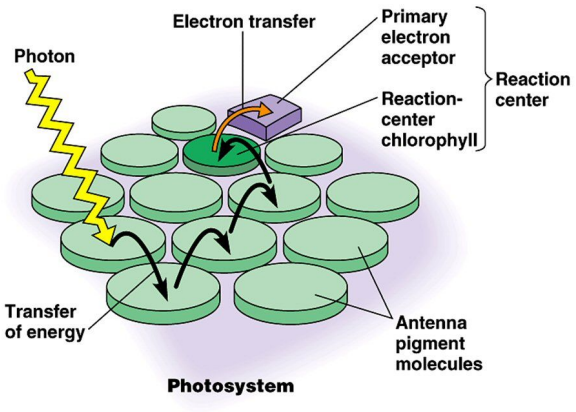



Figure 1: Photosystem with it's antenna and reaciton center.

faster.

Once the energy reaches the reaction center, the wave function collapses to be only on the reaction center. This way all the energy can be harvested for the chemical reactions. However, there is another quantum phenomena that can be harmful for the efficiency of the process: quantum interference.

If the energy is transferred through the pigments as a wave, there might be points where the interference is destructive, losing energy in the way to the reaction center. However, this can be fixed by a factor that all biological systems have: ambient noise.

The photosynthetic antenna complex operates in an intermediate regime between the fully coherent and fully incoherent limits. Counterintuitively, the presence of environmental noise—thermal vibrations, solvent motion, and protein fluctuations—enhances efficiency rather than diminishing it.

This phenomenon, known as environment-assisted quantum transport (ENAQT), arises because moderate levels of decoherence suppress destructive interference patterns that would otherwise trap the exciton in local energy minima. At the same time, coherence is not entirely destroyed; enough remains to preserve the speed advantage of delocalized exploration. The system therefore exploits a finely balanced interplay between coherence and noise, achieving transport efficiencies unattainable in either extreme regime.

The paper Environment-Assisted Quantum Walks in Energy Transfer of Photosynthetic Complexes (2008), written by Masoud Mohseni, Patrick Rebentrost, Seth Lloyd, and Alán Aspuru-Guzik, state that quantum walks are assisted by the environment by the following processes:

The environment will randomly introduce in the system some noise that will collapse the wave function before arriving at the reaction center. This may slow down the process a little bit, but it drastically reduces the probability of suffering from destructive interference.

Other parameters studied in the paper can be seen in figure 2, such as the efficiency of the transfer based on the temperature (a), the amount of noise introduced to the system (also called reorganization energy) (b) and the rate of transfer between pigments. The two lines represent two different pigments at which the experiments were performed (Site 1 pigments for the blue continuous line and site 6 pigments for the red continuous line).

From panel b it can be seen that when no noise is introduced the efficiency of the process is around 15% to 30% lower than the ones observed in nature, and from figure a it can be seen that the efficiency of the site 1 pigments actually increase with the temperature. Usually, the coherence of the systems is reduced when temperature increases, but in this case the results show that for natural conditions (the vertical line in the figure), the efficiency from coherence is quite high.

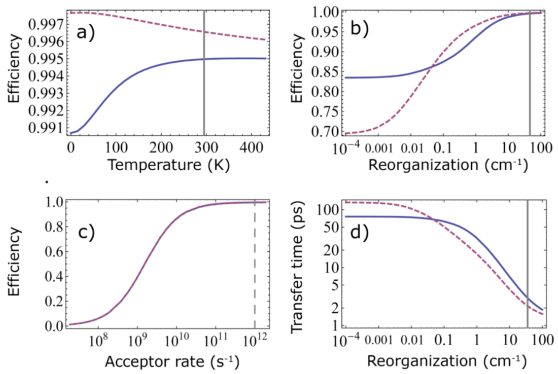



Figure 2: Efficiency of energy transfer process based on temperature, amount of noise and rate of transfer. Source: Mohseni, M. (2008). Environment-Assisted Quantum Walks in Energy Transfer of Photosynthetic Complexes.

## 2.2 Vibronic Coupling

Quantum coherence gives an explanation for the surprisingly high speed of the process, but the high efficiency of the energy transfer is still to be explained. As mentioned earlier, this process has an efficiency of more than 99%, which means that there is practically no energy lost due to dissipation.

Vibronic coupling refers to the mixing of electronic and vibrational degrees of freedom. In photosynthetic pigments, the excited electronic states are strongly modulated by quantized vibrational modes of the surrounding protein scaffold. These vibrations reshape the energy landscape dynamically, periodically bringing donor and acceptor pigments into resonance. When resonance occurs, exciton transfer becomes highly probable, allowing the system to channel excitons downhill in energy toward the reaction center. Importantly, vibronic coupling is not merely a passive consequence of molecular structure; evolution appears to have tuned the stiffness and frequencies of specific vibrational modes so that they assist energy funneling rather than impede it.

A possible explanation would be that the complex structure that pigments form avoid the dissipation of energy in any form and only pass it to the reaction center. However, it was quickly found that this is not the case: they take advantage of the dissipation to direct the flow of energy to the reaction center.

As Jianshu Cao and Robert J. Silbey explain in their paper Optimization of Exciton Trapping in Energy Transfer Processes, the maximum efficiency can be achieved when optimizing dissipative effects through both classical and quantum effects, not avoiding it. The main process through which the system optimizes energy transfer is through vibronic coupling.

In the system there are pigments (groups of chlorophylls) and proteins filling the gaps between pigments (usually called bath). When a group of pigments gets excited by photons they form an exciton, which has a certain electronic state. Vibronic coupling is the process where the electronic state vibrates in resonance with the thermal molecular vibrational modes of the bath.

This resonance allows the energy from the excited state of the electron to flow through thermal dissipation to the next pigment. This way, no energy is lost due to thermal dissipation, as all the energy "lost" from thermal dissipation will be caught by the following pigment, from the high-energy peripheral pigments to the low-energy reaction center.

In Cao's paper, he states that the system is able to maintain the resonance thanks to noise coming from the environment. This would be another example of the importance of the environment in bi-

ological systems, as these are not closed systems and are constantly interacting with their surroundings.

In this paper they also state that the specific structure of the pigments can affect vibronic coupling, so it can be arranged such that it exploits constructive interference in the vibrations.

## 2.3 Quantum tunneling

Once the exciton arrives at the reaction center, its role is to drive charge separation: an electron must be transferred from the excited chlorophyll to the primary electron acceptor within tens of femtoseconds. The protein matrix separating the donor and acceptor imposes a potential barrier too wide for efficient classical transfer. Quantum tunneling enables the electron to bypass this barrier without requiring thermally activated energy, ensuring charge separation occurs before the exciton relaxes. If tunneling were not available, charge separation would be too slow to outcompete recombination processes, and overall photosynthetic efficiency would drop drastically.

Finally in the reaction center, the electron is faced with a wall. Not a literal one, but a mix of water and protein that separates the reaction center to the primary electron acceptor. This wall, if not surpassed rapidly, may represent a huge loss of energy, as the energy carried by the electron would dissipate in the form of heat, and this time there would be no other pigment to accept the thermal energy via vibronic coupling, as the reaction center is the pigment with the lowest potential energy.

This would be the case of the classical approach to the system. If the electron has to overcome the potential barrier only with its own energy, it may have to wait a long time until it has enough thermal energy to pass through the barrier or even worse, it could not penetrate at all.

This is why Nature takes advantage of quantum tunneling to finish the job. The separation between the excited chlorophyll and the pheophytin (the primary electron acceptor) is of around 10 Å. This space is filled with an insulating barrier of proteins, so the electron can not conduct through it. Therefore, the only ways through are with enough energy or exploiting the fact that the wave function of the excited electron extends through the barrier, having a certain probability of spontaneously appearing on the other side.

This field of study, apart from being interesting to know how plants and other living organisms function, can help us get insight into new technologies. If exploited, the knowledge from photosynthesis can be applied to some technologies for energy transferring or energy harvesting, such as solar panels.

Although a lot of progress has been made since the first discovery of quantum effects on photosynthesis, there is still a long way to go, as there are still some inconsistencies between the explanations proposed by researchers and what is observed in laboratories and living organisms. For instance, Jian-Shu Cao mentions in his paper from the year 2020 Quantum biology revisited that Quantum coherence may be insignificant for long-lived energy-transfer processes, but it is still nowadays a big topic of discussion between academics.

Taken together, coherence, vibronic coupling, ENAQT, and tunneling reveal that photosynthesis is not merely a chemical pathway but a highly optimized quantum-dynamical system. The architecture of pigment–protein complexes is arranged so that excitons can exploit quantum delocalization to explore many paths at once, use controlled interactions with the environment to avoid destructive interference, couple to molecular vibrations to direct energy downhill, and finally rely on tunneling to complete rapid charge separation. These coordinated quantum effects explain why photosynthetic energy transfer achieves efficiencies approaching theoretical limits.

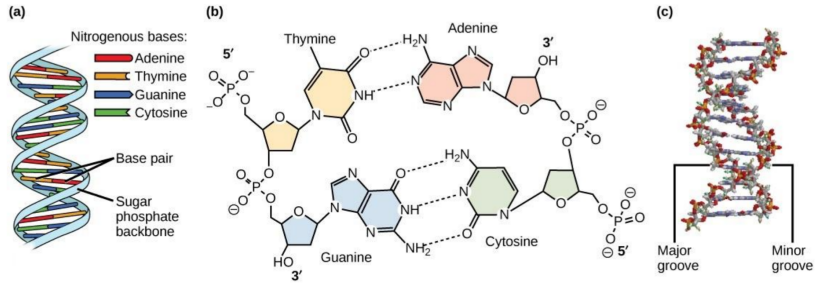



Figure 3: DNA structure and bases.

# 3 Study Case II: DNA mutation

## 3.1 DNA stucture and mutations

Monomers are small, simple molecules that function as building blocks to create bigger molecules. When multiple monomers are bound together, they create a large molecule called a polymer. DNA is a polymer made up by bonding numerous nucleotides, which are monomers. Each nucleotide consists of 3 components: a pentose, a phosphate group and a nitrogenous base. There are four types of nitrogenous bases: Adenine, Thymine, Cytosine and Guanine.

DNA is made up of two strands of nucleotides that are twisted around each other to form a double helix. When the strands come together the nitrogenous bases face each other forming complementary pairs: Adenin- Thymine and Cytosine-Guanine. These base pairs are kept united only by the hydrogen bonds between them; adenine and thymine form two hydrogen bonds and cytosine and guanine form three hydrogen bonds.

A DNA mutation is a random alteration of the nucleotide sequence. It can be caused by environment changes such as chemicals or UV rays or arbitrary mistakes in the DNA duplication process. It is in the latest that quantum effects have been observed.

In broad terms, replication process consists of:

- DNA unwinds at the origin of replication.
- Helicase (an enzyme) separates DNA strands.
- Two RNA are synthesized by copying the complementary bases of each DNA strand.
- DNA polymerase checks and corrects errors to ensure accuracy.
- Each complementary strand, created by the RNA, pairs with the “original” DNA strand.

## 3.2 DNA mutation and quantum effects

The idea that quantum mechanics might play a role in genetic processes has its roots in the mid-20th century. In 1944, Erwin Schrödinger suggested in What Is Life? that heredity must be governed by “quantum laws” because only quantized molecular structures could maintain stability against thermal noise, introducing for the first time the notion that the gene could be a fundamentally quantum object. A decade later, as J.D.Watson and F.Crick constructed the double-helix model, they briefly became confused by textbook illustrations showing rare tautomeric forms of DNA bases. This accidental moment revealed that tautomerism could affect pairing rules, although the mechanism remained purely classical in their interpretation. The first explicit proposal that quantum tunnelling of protons inside DNA could lead to spontaneous mutations was made by Per-Olov Löwdin in the 1950s.

Living cells are not static structures. They exist in an environment physically chaotic: DNA is constantly vibrating, bending, twisting, and interacting with water molecules, ions and proteins. At

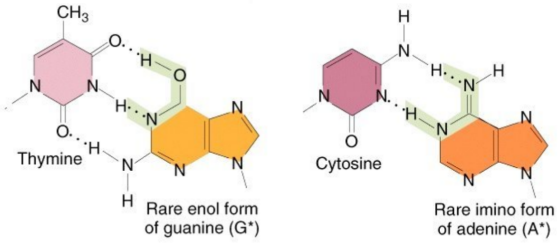



Figure 4: G* pairing with T and A* pairing with C.

the atomic level, Löwdin argued that since protons are extremely light and mobile, they vibrate within the hydrogen bond and can relocate between different bonding positions. In fact, Löwdin proposed that protons' light mass makes tunnelling unavoidable.

When a proton shifts within the H-bond from its usual position, the base changes from its canonical form to its tautomeric form, a rare structural isomer of the nitrogenous base. The canonical bases follow the standard Watson-Crick geometry, whereas, tautomeric bases have different shapes and paring properties: G* pairs with T (instead of C) and A* pairs with C (instead of T).

When the enzyme helicase separates the DNA strands, the H-bond is broken and there is no proton transfer possible, so whatever form the base is in will enter the DNA replication process. Therefore, if a tautomeric pair is separated by the cleavage process before being able to flip back to its canonical form, each tautomer will pair incorrectly i.e. G* pairs with T and C* pairs with A, and RNA will insert the wrong base in the copy strand. Furthermore, the mismatched pair may evade the polymerase's error-checking mechanisms by adopting a nearly indistinguishable geometry from the standard Watson-Crick pair. Thus, resulting in a point-mutation.

Löwdin's hypothesis, nonetheless, soon faded out since it was believed by the scientific community that quantum effects could not survive the decoherent and dissipative cellular environment and therefore, the proton-tautomer state must be extremely short-lived. DNA is immersed in a "messy" environment, physically speaking: molecular vibrations are constantly shaking atoms, temperature is high enough to cause continual thermal noise and ions and water molecules are constantly colliding with its surroundings. All this molecular noise makes it impossible for quantum states to remain coherent or sustained, and causes them to lose their quantum character almost instantly. The collapse of quantum behaviour under environmental disturbance is known as decoherence. The leakage of energy into the surroundings through vibrations, heat, and molecular impacts, is called dissipation.

Together, decoherence and dissipation prevent quantum states from persisting; any quantum-induced shift in the proton's position rapidly becomes undone as the system relaxes back toward its lowest-energy classical configuration. That is why tautomeric forms quickly return to their canonical arrangement—typically within trillionths or even quadrillionths of a second—and no biological effect is produced. At least, that is what was believed at that time.

It was not until the late 1990s and early 2000s that with advances in quantum chemistry (Ab initio and density-functional methods allowed precise calculations of proton transfer potentials, reaction barriers, and tunnelling probabilities within DNA base pairs) that it was shown that tautomerisation is chemically possible and that tunnelling contributes to proton transfer more than previously appreciated.

The behaviour of DNA Hydrogen-bond protons is fundamentally quantum mechanical: each proton exists as a wavefunction, a delocalized probability cloud capable of spreading, oscillating, superposing, and—critically—tunneling through energy barriers, allowing it to extend into regions that would classically be forbidden. In DNA base pairs, the proton responsible for stabilizing the hydrogen bond experiences a double-well potential: one well corresponds to the canonical position and the other to the tautomeric position, with an energy barrier separating them. Classically, the proton would remain

confined to the canonical well because it lacks the thermal energy needed to climb over the barrier. However, according to quantum mechanics, its wavefunction can penetrate and extend through the barrier, giving the proton a finite probability of appearing in the tautomeric well. The shape of the double-well potential—its depth, width, and asymmetry—determines how easily tunnelling occurs and how stable the resulting tautomeric state can be.

In 2021, Slocombe, Al-Khalili and Sacchi published a paper that covered whether quantum tunnelling of protons in DNA base pairs could occur frequently enough to influence biological mutation. To address this, the authors first constructed accurate potential energy surfaces for proton transfer in adenine–thymine and guanine–cytosine pairs using advanced quantum-chemical calculation. These surfaces revealed energy barriers far higher than the thermal energy available in cells, indicating that classical over-the-barrier proton transfer is extremely unlikely and that proton transfer occurring on biologically relevant timescales must be dominated by quantum tunnelling.

Treating the proton as a quantum particle in this double-well potential, the authors calculated tunnelling probabilities as a function of vibrational energy and found that tunnelling probability increases sharply with proton energy. Then, by invoking a Boltzmann distribution, the authors estimated how frequently the proton occupies higher-energy vibrational states under thermal equilibrium. Although most of the proton population occupies low-energy states, a small fraction is thermally promoted to higher-energy quasi-bound states. Therefore, if thermal vibrations populate higher vibrational states, then tunnelling will be enhanced.

The authors the applied this framework to DNA base pairs and studied inter-base proton transfer (what takes place inside the DNA) and found out that the Adenine–thymine proton transfer has a forward barrier is of 0.58 eV, a reverse barrier of 0.01 eV, which is extremely small, and a tautomer lifetime of $10^{-14}$ s. That proves that tunneling can create A*–T*, but the tautomer collapses back almost instantly. On the other hand however, for guanine–cytosine proton transfer, the forward barrier is 0.69 eV, the reverse barrier is 0.27 eV and the tautomer lifetime is $10^{-14}$ s. Although, $10^{-14}$ s is still short, it is 1000× longer than A–T tautomer lifetimes and long enough that a small fraction may survive through the early stages of replication. Therefore, it was concluded that A–T tunneling is not a biologically relevant source of spontaneous mutation but G–C quantum tunneling can contribute to spontaneous point mutations.

Moreover, it was also examined intra-base proton transfer (within single bases) because during the replication process bases are temporarily single-stranded. It was discovered that within isolated bases the energy barriers were of 1.6-2.1eV, much larger than in the inter-base case, nevertheless, the tautomers produced lasted much longer, about $10^6$ - $10^{14}$ second. Therefore, although these tautomers are very unlikely to be produced by tunneling, they are extremely stable. This study had been conducted treating the proton as effectively isolated from its surroundings treating thermal motion as a background statistical assumption rather than a physical dynamical process. However, it clearly showed that population of thermally excited vibrational states is essential for enabling proton tunnelling, so the authors concluded that a full open quantum system (OQS) approach, that is, taking into account the environment, would be required. That became the focus of their subsequent 2022 study.

The 2022 paper introduced for the first time a model that treated the hydrogen-bond proton inside the DNA as an open quantum particle constantly interacting with its environment. This is achieved mathematically through a Gorini–Kossakowski–Sudarshan–Lindblad master equation, the standard framework for describing quantum systems interacting with an external thermal bath. In this formulation, the proton evolves according to both quantum unitary dynamics and environmentally induced transitions between vibrational eigenstates. Such a model acknowledges that biological environments do not simply destroy quantum effects but can also feed energy into the system. Knowing from their previous study that A-T equilibrium is negligible regarding DNA mutations, the authors only focus on the G-C equilibrium.

Slocombe, Al-Khalili and Sacchi assumed, as in agreement with scientific literature, that the two protons undergo an asynchronous transfer. Suggesting that the rate-determining process is the trans-

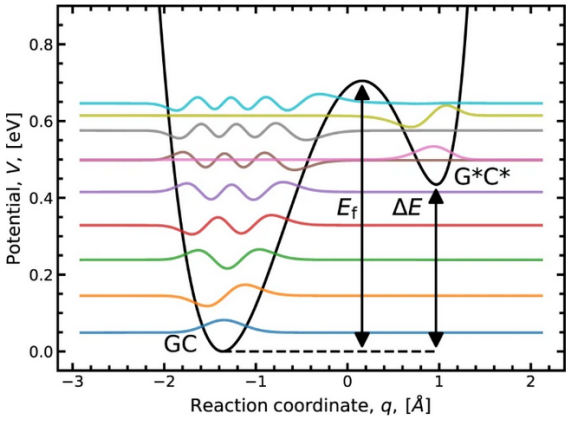



Figure 5: Potential energy with respect the reaction coordinate.

fer of a single proton across the barrier, with the other proton then moving in the opposite direction to preserve electroneutrality. The thermodynamic equilibrium constant, thus, can be written as: $K_{eq-qm} = \Gamma_{qm}[G^*\text{–}C^*]/[G\text{–}C]$. Where $[G*\text{–}C*]/[G\text{–}C]$ is the ratio between the concentration of products (tautomeric pair) and reactants (canonical pair) present at chemical equilibrium and $\Gamma_{qm}$ is exclusively a quantum activity coefficient containing effects such as thermally enhanced tunnelling, zero-point energy, decoherence and dissipation. Keq can be obtained either from the difference in Gibbs free energy, $\Delta G$, between products and reactants - via $K_{eq} = \exp(G/K_bT)$ and calculated with ab initio methods, or measured experimentally. However, without an accurate estimate of the activity coefficient, it was not possible to determine the quantity of interest $[G^*\text{–}C^*]/[G\text{–}C]$. Instead, the authors carried out a fully quantum mechanical calculation to obtain the $G^*\text{–}C^*$ occupation probability within an open quantum system (OQS) approach.

The authors plotted the proton transfer process by projecting the reaction pathway on a generalised reaction coordinate, that connects the canonical to the tautomeric state via a single transition state, on a chemically accurate potential energy surface (PES). The reaction is characterised by an $\Delta E$ (energetic asymmetry between the bottoms of the two potential wells) of 0.435eV, a high forward barrier of 0.705eV and a reverse barrier of 0.270eV.

When solving the Schrödinger equation for the first 10 quantum energy levels, it was found that the first six eigenstates are confined entirely to canonical well, giving them negligible amplitude in the tautomeric region, and it was not until the seventh of them that there is a non-zero amplitude in the tautomeric well, effectively increasing the chances of finding the proton on the classically forbidden side of the double-well. This is the central mechanism through which the authors demonstrate that thermal noise, instead of suppressing tunnelling, actively enhances it by driving the proton into vibrational states that facilitate barrier penetration. The model therefore demonstrates a counterintuitive interplay: biological noise, while destroying long-lived coherence, simultaneously boosts the proton into energy regimes where tunnelling becomes more effective.

The writers then proceeded on determining the lifetime of the tautomeric state. The proton's dynamics were described using: the Caldeira–Leggett quantum Brownian motion equation, in which the quantum system, a proton in the double-well potential, interacts with a surrounding infinite thermal bath of harmonic oscillators, representing the cellular environment. The equivalent phase-space formulation of the Caldeira and Leggett master equation, also known as Wigner–Moyal–Caldeira–Leggett (WM–CL) equation, is written as:

$$\frac{\partial W}{\partial t} = -\frac{p}{m}\frac{\partial W}{\partial q} + \frac{\partial V}{\partial q}\frac{\partial W}{\partial p} - \frac{\hbar^2}{24}\frac{\partial^3 V}{\partial q^3}\frac{\partial^3 W}{\partial p^3} + O(\hbar^4) + \gamma\frac{\partial pW}{\partial p} + \gamma m k_B T\frac{\partial^2 W}{\partial p^2}.$$

The quantum contribution Keq-qm was calculated by monitoring how the flux of the probability changes between the left and right-hand well and was found to be of the order of $10^5$, indicating that the quantum contribution to the reaction rate is non-trivial. With this value, it was obtained a forward

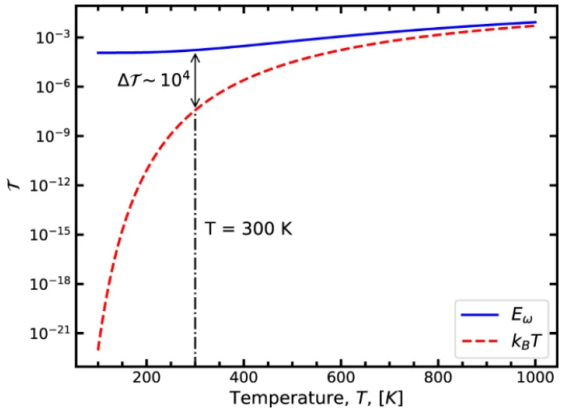



Figure 6: Energy in funciton of the temperature.

and reverse reaction rate of $k_f = 7.6110^5 s^{-1}$ and $k_r = 1.6910^{13} s^{-1}$, respectively, whereas the half-life of the reactant and product is now $9.1110^{-7}$ and $4.0910^{-14} s$, respectively. Comparatively, setting the quantum contribution ($K_{eq}$) to unity and considering a classical equilibrium constant, $K_{eq}$, of the order of 10-8, as in agreement with previous studies, it was obtained a classical forward and reverse reaction rate of $k_f = 7.596 s^{-1}$ and $k_r = 1.692108 s^{-1}$. Whilst, the half-lifes of the reactant and product are 0.0913 and $4.110^{-9} s$, respectively.

From the results above it can be corroborated that quantum corrected half-lives are significantly shorter than the classical estimates, in fact, tunnelling increases proton transfer by 5 orders of magnitude, indicating that protons are sufficiently delocalised along the hydrogen bond and populate both states. Furthermore, the forward and reverse reaction rates are so quick that the system does not remain in a single configuration but constantly fluctuates between the canonical and tautomeric forms. This fast interconversion timescale suggests that an equilibrium description can be safely adopted, that is, the ensemble of DNA molecules always contains a constant population of tautomers.

With these recent insights, the authors solved the OQS to equilibrium at 300K, biological temperature, and computed a tautomeric occupation probability of $1.73 \times 10^{-4} s$. In contrast to what classically had been predicted, according to the increment of energy between the canonical and tautomeric states ($\Delta E = 0.435 eV$, see Fig. 6) the classical thermal equilibrium constant for the model system would have to be $K_{eq} = \exp(-(0.435 ev)/K_b T) = 5.410^{-8}$. As it can be observed, there is a factor difference of $10^4$ between the thermalised tautomer occupation probability obtained from the two treatments.

Thus, it is not the lifetime of a single tautomer that matters, but it is the equilibrium ratio between canonical and tautomeric forms inside the DNA that determine the mutation probability. In other words, the mutation probability depends entirely on the equilibrium tautomer population.

In conclusion, the 2022 open-quantum-system model shows that proton tunnelling in guanine–cytosine base pairs arises naturally from the interaction between the hydrogen-bond proton and its surrounding thermal environment, and that this quantum mechanism allows proton transfer to occur far more readily than classical, over-the-barrier theories would permit. By explicitly incorporating environmental coupling, decoherence, and dissipation, the model demonstrates that thermal fluctuations actively populate higher vibrational states whose wavefunctions extend across the proton-transfer barrier, leading to a small but significant tautomeric population under physiological conditions. Because tautomeric bases are known to mispair during DNA replication, these results suggest that quantum tunnelling may contribute to spontaneous point mutations. While the study does not demonstrate this process directly in vivo, its calculations provide a quantitative and physically grounded basis for reconsidering proton tunnelling as a plausible intrinsic factor in genetic variation, and thus motivate future experimental investigations aimed at testing this mechanism in biological systems.

# 4 Study Case III: Enzyme Cathalysis

Enzymes are biological catalysts, mostly proteins, that achieve extraordinarily high reaction rates by providing alternative pathways that are faster and require lower activation energies. These accelerations, which can exceed the order of 10 to the power of 17 (Radzicka & Wolfenden, 1995), are paramount to life as they enable metabolic processes, DNA replication and other essential processes to occur within the strict time constraints of a living cell. However, this proficient catalytic performance's explanation lies beyond the classical framework of Transition State Theory (TST) (Kohen & Klinman, 1999). A complete understanding requires a quantum mechanistic approach. Classically, enzymes are understood to lower the activation energy by stabilizing the transition state. Yet, for reactions involving hydrogen transfer the stabilization of the transition state is not enough to account for the large transition rates or catalysis at cryogenic temperatures. These discrepancies compel us to consider the wave-like nature of atoms, where particles can go over energy barriers or tunnel through them.

Hydrogen transfer is highly relevant to the transfer of reducing equivalents (species that transfer the equivalent of one electron in a redox reaction) between substrates (molecules upon which enzymes act) and cofactors. Many enzymes, especially those that catalyse redox reactions, have two compounds: the apoenzyme (the protein) and the cofactor which is not a protein but is, nonetheless, essential for catalytic activity to occur. The cofactor can be an inorganic or organic compound. When it is an organic compound it is called coenzyme (Solomon et al., 2007).

Hydrogen transfer in enzyme-catalyzed reactions like all other enzyme-catalyzed reactions had been classically studied through the lens of transition-state theory (TST) (Kohen & Klinman, 1999). The relevance of quantum-mechanical tunneling of protons in chemical reactions was firstly hypothesized around the 1950s. In his 1959 book "The proton in chemistry" Ronald Percy Bell explained that proton tunneling might be significant in enzyme-catalyzed reactions. This set the ground for a new field of study: non-classical enzymology.

Later on, in 1974, John Joseph Hopfield published "Electron transfer between biological molecules by thermally activated tunneling" in which the 2024 physics Nobel laureate described a quantitative model that explained how electrons can transfer so effectively between biomolecules through relatively large distances and at physiological temperatures. Some of the biological examples mentioned in this paper are photosynthesis, electron transport chains and redox enzymes. Non-classical hydrogen transfer in enzyme-catalyzed reactions has been empirically shown to occur ever since by different scientists like J.P. Klinman, A. Kohen or M.J. Knapp.

The conventional theory studied enzyme-catalyzed reactions through transition-state theory. In this theory reactions occur when its activation barrier is overcome (Kohen & Klinman, 1999). However, it couldn't account for accelerations of the reaction of the order of 10 to the 17 when using enzymatic versus non-enzymatic reactions (Radzicka & Wolfenden, 1995), anomalous isotope effects and activity at low temperatures (Roston, Islam & Kohen, 2013).

An example of this acceleration of the reaction was found in 1995 and published in Science magazine in an article called "A proficient enzyme" by Radzicka and Wolfenden. Radzicka and Wolfenden showed that ODCase (orotidine 5'-phosphate decarboxylase) accelerates the decarboxylation reaction by approximately 17 orders of magnitude compared to the uncatalyzed reaction in water. The reaction studied was the decarboxylation of Orotidine 5'-monophosphate (OMP) to uridine 5'-monophosphate (UMP), where a mol of OMP produces a mol of UMP and a mol of $CO_2$. UMP is relevant as it is the precursor for pyrimidine nucleotides (Moffat & Ashihara, 2002) that form the two polynucleotide chains intertwined in a double helix shape of DNA (Solomon et al., 2007). This catalytic proficiency defies a purely classical TST explanation because the active site cannot lower the energy barrier enough to account for such a large enhancement. This discrepancy points to a non-classical alternative and it is, therefore, compelling to take quantum mechanics into consideration. ODCase's exact mechanism is complex and its details are not yet fully understood. However, it is well known that charge delocalization in the transition state and proton tunneling contribute significantly to its extraordinary proficiency.

The quantum mechanics framework needed to approach an explanation on the functioning of en-

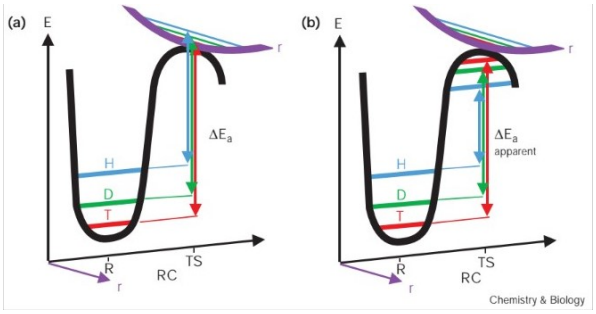



Figure 7: Protium (blue), deuterium (green) and tritium's (red) activation energy. (a) following the semiclassical model. (b) adding a tunneling correction for the TST model. source: Klinman, J., & Kohen, A. (1999). Hydrogen tunneling in biology. Elsevier Science.

zymes has an underlying limit intrinsic to nature itself, the uncertainty principle. The uncertainty principle prevents the lowest energy a particle can occupy from being the electronic potential minimum, which is called the zero point energy (ZPE). For smaller and less massive particles the ZPE is higher and the difference between ZPEs of ground states and transition states is reduced. This phenomenon contributes to the Kinetic isotope effects (KIEs) which reflect the different behaviours of particles of different mass along the same electronic potential surface. The theory of KIEs is considered a semiclassical theory as it corrects TST whilst ignoring the uncertainty in the reaction coordinate close to the transition state.

Hydrogen's KIEs are especially useful as the mass of deuterium and tritium are proportionally much larger than protium's. This results in particularly significant differences in the size of the energy barrier between the ZPEs of the ground state and the transition state and at the same time increases the KIEs . Figure 7a shows these semiclassical effects for the hydrogen isotopes (Klinman, 1999). Moreover, lighter isotopes are able to tunnel through larger distances, that is to say broader barriers, compared to their heavier counterparts (Klinman & Kohen, 2013).

On the other hand, the uncertainty in the position of the transferred particle along the reaction coordinate can be estimated with the de Broglie wavelength. This introduces the tunneling correction which also takes into account that at a certain energy level the wavelength of the particle exceeds the barrier's width and the product is formed without having needed to reach the transition state's energy. The differences between the activation energies of the different hydrogen isotopes are even larger. Figure 7b shows the semiclassical effects that can also be seen in Figure 7a paired with this tunneling correction. It is crucial to note that even though all of these models regard barriers as static, energetic barriers fluctuate.

All of these phenomena can be quantified. The Bell correction presents several modifications to Arrhenius's kinetic theory in order to account for quantum tunneling. In this model, the reaction rate constant ($k$) can be found as a product of the tunneling transmission coefficient ($\kappa_{tun}$) and the reaction rate constant following TST ($k_{TST}$). The tunneling transmission coefficient ($\kappa_{tun}$), however, can be larger than one as it is defined as the sum of a constant smaller than one plus the reaction rate of tunneling ($k_{tun}$) over the reaction rate following TST($k_{TST}$). This second term, representing the tunneling flux, can be extremely large and that is what leads to accelerations as the ones previously mentioned (Knapp & Klinman, 2002). The differences between the reaction rate constant ($k$) and the reaction rate constant only taking into account TST ($k_{TST}$) are numerically what conceptually differs from Figure 7a and Figure 7b. However, Bell's model still largely treats the energy barrier as static. Modern enzymology is revolutionary because it posits that enzymes are dynamic machines that constantly shape the energy barrier.

The validity of this quantum framework is demonstrated by its ability to explain the empirical indicators of a non-classical theory that defy classical analysis. This is evidenced, first, by the kinetic isotope effect ($k_H/k_D$) for hydrogen transfer reactions, which is consistently observed to surpass the

classical limit of  7 (Kohen & Klinman, 1999). Furthermore, hydrogen-transfer reactions often exhibit a weak or even inverse temperature dependence of the reaction rate and the KIEs. Equally telling is the Swain-Schaad exponent, which relates protium over tritium and deuterium over tritium KIEs, which consistently exceeds its classical limit of  3,3. The most conclusive evidence, however, comes from the systematic deviations from classical Arrhenius behaviour. In the Arrhenius equation, the reaction rate depends on the activation energy ($E_a$) and a pre-exponential factor (A). Another factor that indicates that this framework has broken down when experimenting with tunneling in a reaction is that Arrhenius plots become non-linear and the apparent A and $E_a$ become strong functions of temperature (Knapp & Klinman, 2002). The observation of these anomalies not only confirms tunneling, but it also requires an explanation for how such efficiency in quantum barrier crossing is achieved by enzymes. This points towards the enzyme being a dynamic quantum machine.

The enzyme Soybean lipoxygenase-1 (SLO-1) synthesizes fatty acid hydroperoxide at 1,4-pentadienyl positions (Vick & Zimmerman, 1987) and provides a definitive case study. It catalyzes a C-H bond cleavage with a KIE of  80 at room temperature. It is also notable that the KIE are so high that, in 1994, C. C. Hwang and C. B. Grissom conducted a study to rule out the possibility that these high KIEs were due to magnetic effects. This combination of a large KIE with negligible activation energy difference for protium versus deuterium transfer is considered to clearly show that the reaction rate is dominated by the quantum tunneling probability rather than classical thermal activation (Knapp & Klinman, 2002).

Quantum mechanics/molecular mechanics (QM/MM) simulations provide an atomic-scale view of this dynamic quantum machine. This computational technique allows the reacting atom to be studied through the lens of quantum mechanics, therefore enabling the visualization of bond-cleavage or formation and tunneling. At the same time, however, the surrounding protein and solvent are treated with classical molecular mechanics. These simulations track the reaction dynamics over time so as to demonstrate that proteins compress the donor-acceptor distance, which narrows the width of the tunneling barrier. The probability density of the transferring hydrogen can dramatically increase by delocalizing across the barrier. (Hwang & Warshel, 1996). Studies that employ QM/MM have hugely contributed to the knowledge on the geometry and electronic structure of the tunneling-ready state (TRS), which is the exact state that enables the tunneling to occur (Roston, Islam & Kohen, 2013). QM/MM have, therefore, transformed a theoretical concept into a quantitatively observable phenomenon.

The transfer of a hydrogen from its reactant state to its product state can be interpreted as a double well system where the probability of finding the particle at the reactant and product wells overlap and, therefore, the hydrogen can tunnel itself from one of the wells to the other (Kohen & Klinman, 1999). How probable tunneling is mainly depends on the barrier's width and not so much on its height. The narrower the barrier the lower the energy of a particle has to be for it to be able to tunnel and the more frequent tunneling will be (Hay et al., 2010).

Symmetry is another factor of paramount importance when regarding tunneling probabilities. A perfectly symmetrical double-well system will cause the particle to bounce back and forth between the reactants' and the products' well. This will mean that, even if the reaction occurs, it can be reversed at any time. The loss of coherency eliminates this reversibility and stabilizes the products (Kohen & Klinman, 1999).

For a reaction to efficiently facilitate a net chemical transformation the enzyme needs to transiently achieve several optimal conditions, more specifically the barrier needs to be as narrow as possible and the energy levels of reactants and products need to match. This optimized geometry is termed tunneling-ready state (TRS) in Marcus-like models. The enzyme accomplishes TRS through thermally activated movements that are inherent to the protein's architecture; this process is called vibrationally enhanced tunneling. These movements are often low-frequency motions or localized vibrations that compress the distance between the hydrogen donor and acceptor atoms for spans of the order of femto or picoseconds, effectively "squeezing" the potential energy barrier. The TRS formation is paramount as it exponentially increases the tunneling probability, thus providing enzymes with their proficiency.

Overall, enzyme evolution has led to extremely efficient compounds that exploit quantum mechanical properties of matter to enhance reaction rates. They have architectures that can promote or prevent certain quantum phenomena from happening by modifying their dynamical behaviours. Were it not for hydrogen quantum tunneling, reactions wouldn't be fast enough for life to be sustainable. However, it is remarkable to note that hydrogen quantum tunneling is not an ubiquitous phenomenon. This powerful catalytic strategy is thus reaction-specific and highly exploited for particularly kinetically challenging steps. This insight opens new pathways in fields like drug design or enzyme engineering, where mimicking these quantum-optimized dynamics represents a key frontier.

# 5 Conclusions

The investigation into photosynthesis, DNA mutation, and enzyme catalysis reveals that life is not merely a complex chemical pathway but a highly optimized quantum-dynamical system. Across these diverse biological processes, quantum mechanics provides the explanatory power required to reconcile the inconsistencies found in classical models.

In the case of photosynthesis, the architecture of pigment-protein complexes allows excitons to exploit quantum delocalization. By existing in a superposition of states, energy can explore multiple paths simultaneously, avoiding the "random walk" inefficiencies predicted by classical physics. Crucially, the system utilizes quantum tunneling to bypass potential barriers for rapid charge separation, preventing energy dissipation.

Similarly, the study of enzymes, such as ODCase, demonstrates that quantum tunneling is essential for overcoming reaction barriers. The catalytic acceleration observed—up to 17 orders of magnitude-cannot be explained by classical Transition State Theory alone. Tunneling allows particles to penetrate energy barriers rather than requiring the thermal energy to climb over them, a mechanism that is also central to spontaneous genetic variation. In DNA, the quantum tunneling of protons between base pairs allows for the formation of tautomers (G*-C*), which, while short-lived, can lead to point mutations if stabilized during replication.

Perhaps the most profound insight across these studies is the role of the biological environment. Contrary to early beliefs that thermal noise would destroy quantum effects (decoherence), recent models suggest that biological systems function as Open Quantum Systems (OQS). In both photosynthesis and DNA proton transfer, environmental noise does not hinder the process but actively assists it. Through mechanisms like Environment-Assisted Quantum Transport (ENAQT) and vibronic coupling, thermal fluctuations suppress destructive interference and populate vibrational states that enhance tunneling probabilities.

Ultimately, understanding these quantum biological mechanisms offers more than just theoretical insight; it holds the key to future technological innovation. If we can replicate the quantum strategies employed by photosynthetic complexes, we may revolutionize energy harvesting technologies, such as solar panels, achieving efficiencies previously thought impossible.

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